Sri Lanka Institute of Information Technology

IT1040 - Fundamentals of Computing

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Automated Health Care Monitoring System

Proposal Document

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1. Background

Being healthy is the key to living a rewarding and effective life. Regular health monitoring is essential for good health all around the world. So, healthcare monitoring systems have progressively assumed this responsibility from past to the present and future. [1]

Modern healthcare monitoring systems have undergone significant advancement. They can be manual manpowered systems and automated, technological systems.

The heart rate (HR), blood oxygen saturation (Spo_2), and body temperature are three of the most important indicators of an individual's cardiovascular and respiratory health. [2]

Heart rate monitoring

Heart rate monitoring involves tracking and measuring a person's heart rate, often using sensors and electrodes to detect changes in heart activity. [4]

Current heart rate monitoring systems.

- Wearable Devices (smart watches, fitness bands)
- Chest Strap Monitors
- ECG/EKG Monitors
- Smart Clothing
- Implantable Devices

Blood oxygen saturation monitoring

Blood oxygen saturation monitoring involves measuring a person's percentage of oxygen in the blood. [3]

Current blood oxygen saturation monitoring systems.

- Pulse oximeters
- Wearable Devices
- Medical Monitors
- Smart Phones

Healthcare professionals have shared both positive and negative opinions about these healthcare monitoring systems. They analyze patient engagement, data privacy, availability.

A wide variety of diseases, disorders, and health conditions can be treated or supported through automated healthcare monitoring systems. This is why the healthcare field continually demands more advanced and enhanced versions of these systems. [1]

2. Problems and Motivation

2.1 Problem Statement

Traditional psychological tests such as surveys, questionnaires, and interviews are still used extensively in mental health monitoring today. Despite their value, these approaches pose serious difficulties for both patients and mental health professionals.

They Take a Long Time

It takes ages to get a full report on someone's mental state using these old methods. If a doctor sees the same patient often, this really slows everything down.

It's Hard to Write Down Information

Another problem is that doctors have to write down and keep all patient information by hand. It's difficult to save these reports, and then it's hard to look at all that "talking" information to decide what to do next. This also makes it tough to see if a patient is getting better.

New Machines Cost Too Much

There are new ways to check mental health using special machines, but these machines are super expensive. Most doctors and patients can't buy them, so they're not used very much.

Machines Can Scare People

For many people with mental health issues, being hooked up to modern machines can feel scary or stressful. This might make them not want to get treatment. We need machines that are simple, work well, and don't cost a lot.

2.2 Motivation

The creation of an automated healthcare monitoring system that includes blood oxygen saturation (SpO2) and heart rate monitoring devices will benefit society in many ways, including patients and medical professionals. [2]

Enhanced Data Management for Health Professionals

Heart rate, SpO2, and body temperature readings are among the patient data that doctors and psychologists will be able to organize and store. This will make analysis simple and enable them to track the exact development of patients' treatments over time. Future prescriptions and treatment modifications can be made more effectively and precisely with the help of easily accessible historical data.

Early Detection of Mental Health Conditions through Physiological Indicators

The system can offer early insights into a user's stress levels or other physiological responses suggestive of changes in mental health by continuously monitoring physiological indicators such as heart rate, SpO2, and body temperature. [2] In order to help users manage their condition, this system can then provide recommendations or alerts. This can be very helpful in identifying and treating long-term mental health issues early on, possibly even before they worsen.

Continuous and Real-Time Health Monitoring

Through a real-time web dashboard, medical professionals will be able to monitor their patients' physical and mental health. Better physiological data visualization (heart rate, SpO2, and temperature) will be possible with this dashboard, allowing for ongoing monitoring and prompt action when needed.

3. Aims and Objectives

The healthcare industry has been getting advanced scientifically and technologically through wireless-sensing devices. Automation of the healthcare system with the internet of things has been implemented and attains accurate results. [6]

3.1 Aim

The aim of this study was to provide health monitoring for elderly and disabled persons at home by using fully automated signal measurement with personal identification to support daily healthcare and improve quality of life. The “OxyBeat” is to scan and display real time data on patient’s heartbeat, blood oxygen saturation (SpO2), and body temperature and a timeline of them index through a wearable device. A set of proper sensors and a micro-controller will be used for data collecting and analysis. Analyzed information and data will be accessible on a web dashboard to timeline of them.

3.2 Objectives

3.2.1 Main Objectives

1. Enhance Patient Safety

Identify potential health risks and prevent adverse events. Identifying the patient's condition and providing necessary advice for the patient's well-being through timely notes.

2. Interoperability

Ensuring seamless integration with existing healthcare systems and devices. This makes it easier for the patient to connect with doctors and receive the treatment and advice they need.

3. Algorithms and Data processing

Providing a real-time data graph so that care providers and patients can gain proper knowledge by comparing the data provided by the sensor with the patient's subsequent data.

4. Reduce hospitalizations

The timely reports and data provided through the website can also improve the patient's health, thereby reducing hospitalizations and making it easier to check the patient's condition before hospitalization.

5. Mobile dashboard development

Creating a user-friendly interface for healthcare professionals to display pulse rate, blood oxygen level, body temperature real-time data comparison index, and significant changes in those parameters.

3.2.2 Assigned Objectives for Members

TABLE I
Assigned Objectives for Members

IT Number	Name	Assigned Objective	Objectives Description
IT25102047	Palihakkara P. J.	Historical data graph	Create historical data graphs in mobile interface using the readings from the sensors.
IT25102048	Pathirana G. M. S.	Buzzer indicator	Indicate normal and abnormal readings through buzzer.
IT25102049	Minupama S. G. Y.	LED indicators	Indicate normal and abnormal readings through 3 green and red LED pairs
IT25102050	Bingusara K. P. D	Relaxed/ Stressed indicator	Create an indicator in the mobile interface that indicates whether the user is relaxed or not, based on whether all the readings are good or bad.
IT25102051	Randeeth W. A. V.	Colour coded dial indicator	Create colour coded dial indicators in mobile interface for each reading
IT25102054	Dinali L. D. G. N.	Real-time numerical readings	Create real-time numerical readings indicators for each reading in mobile interface.

4. System Diagram

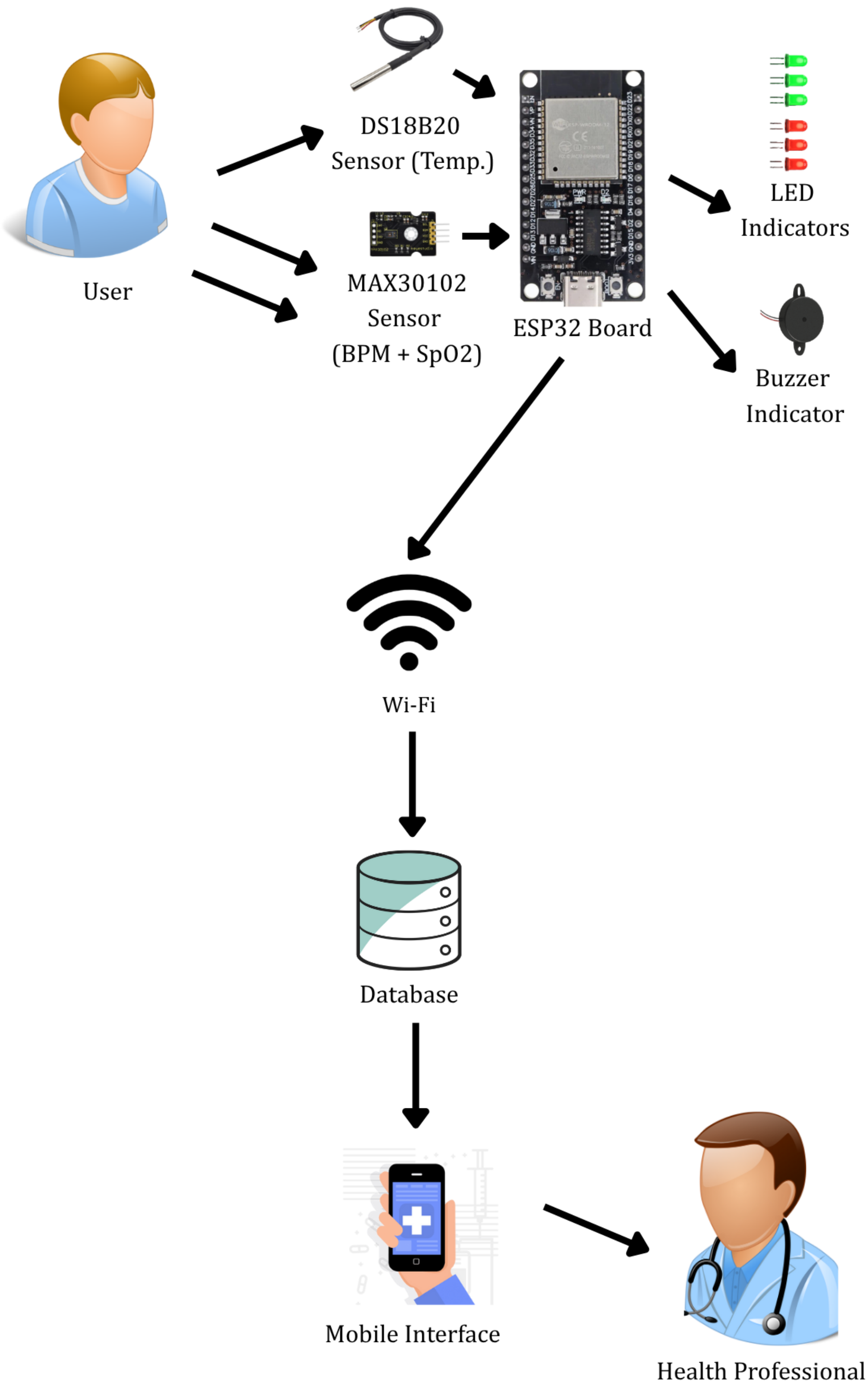

Fig. 1. System diagram

5. Methodology

5.1 System Overview

The "OxyBeat" system is a portable device that continuously monitors heart rate (BPM), blood oxygen saturation (SpO2), and body temperature. This section will describe the techniques and technologies utilized in the development of this system. It is starting with hardware selection and progressing to software applications and data presentation.

5.2 Hardware Components

We require a set of hardware that will do a specific task for our "OxyBeat" project.

1. MAX30102 and DS18B20 Sensors

These sensors will measure BPM, SpO2 and body temperature accurately. That is why we choose these sensors. [5]

2. ESP32 Microcontroller

This ESP32 development board will give us powerful processing and built-in Wi-Fi capabilities that will make smooth data transferring to our mobile interface. We will use Arduino IDE to program it. [6]

3. Physical LEDs

The user can get immediate visual feedback from this dedicated pair of LEDs. Green-colored LEDs will indicate normal readings, while red-colored LEDs will indicate abnormal readings.

4. LiPo Battery (3.7V) and TP4056 LiPo Charger Module

This LiPo battery and its charger module will power our system and provide it with extreme portability that provides ease of use to the user. [8]

5. Breadboard and Jumper Wires

We will use a breadboard and jumper wires because they will give ease for quick testing and assembly.

7. 220-ohm Resistors and 4.7k-ohm Resistor

220-ohm resistors will protect LEDs 4.7k Ohm resistor will protect DS18B20 sensor from high voltage.

8. Micro USB Cable

We will use this for programming and charging.

9. Plastic Enclosure

A small plastic enclosure will protect our electronics and will give our product a finished look.

5.3 Communication Module

- **Wi-Fi:** By using built-in Wi-Fi, ESP32 will connect to a local network that will transfer processed data into Blynk cloud. [6]

5.4 Data Processing

- ESP32 will collect data that the user inputs to the MAX30102 and DS18B20 sensors.
- Firmware in this system will process that data and will give an accurate heart rate, blood oxygen saturation, and body temperature.
- BPM 60-100, SpO_2 95%-100% and temperature 36.1°C to 37.2°C ranges will be indicated as normal readings, and other readings will be indicated as abnormal readings in both physical LEDs and the mobile interface. [3], [4]

5.5 Software Development

- **Arduino IDE (C/C++):** We will use Arduino IDE as our main environment for writing firmware for ESP32.

5.6 Mobile Application

- **Blynk Application:** We will use the Blynk application for remote monitoring and data visualization, and its dashboard will show real-time BPM, SpO2 and body temperature levels, graph previous data, and send critical alerts via push notification. [7]

6. Testing and Evolution

Testing & evolution is a methodical procedure used to the combined system is accurate works correctly and reliable. [9]

This automated healthcare monitoring system incorporates heart rate monitoring system, blood oxygen saturation monitoring system and body temperature monitoring system.

The SP02, body temperature and heart rate monitoring features work together smoothly and at the same time without getting each other's way.

6.1 Unit Testing

This is basically the process of testing individual components separately. In the system the pulse oximeter sensor and body temperature sensor are basically tested for appropriate connection and accurate reading for SP02, HR body temperature. ESP32 microcontroller basically checked for correct input reading from sensors and for proper data transmission to web interface.

6.2 Integration Testing

This test verify the hardware and software functioning properly. In this case SP02, heart rate and body temprature. Sensors smoothly Send their information to the ESP32 microcontroller. It guarantee that the live data from both sensors is correctly handled & saved. and Shown on the website.

6.3 System Testing:

System tests the functionality of the entire system under actual use condition. In this combined system, we test,

- The SP02 sensor continuously measures oxygen saturation, pulse rate and body temperature and
- The system provides warning if the user’s readings are abnorma.

6.4 Performance Testing:

This process ensure how the system operates, for extended period and collects information from many different sensors at the same time.

6.5 Usability Testing

This check how easy and comfortable to use the system.

- Check whether when you wear heart rate, SP02 and temperature sensors
- Is the dashboard easy to use.

6.6 Reliability and Fault Tolerance

- This stage checks if the system can still work well even when things go wrong or are not normal conditions such as: sensor disconnection, poor Wi-Fi signal and temporary power interruption are involve putting the system in tough situations on purpose.
- After the checking process it gets back to normal and makes sure all the information stage correct and incomplete.
- The ESP32 ability to reconnect to the network and resume sensor data collection without user intervention is also tested.

- Backup logging on local memory or buffer mechanism is verified to ensure critical health data is lost.

6.7 Validation and final review

- In this phase the final phase the system is validated against initial design requirement.
- SP02, heart rate, and temperature reading are compared with certificated medical devices for accuracy. [3], [9]
- Clinical validation by health professional adds credibility to all collected data.
- In usability stage review ensure that the system is practical for daily use.

All collected testing data & performance logs, user feedbacks and error report helps to make "final report" and this report help to further improvements.

7. Proposal Budget

TABLE II
PROPOSAL BUDGET

Item	Quantity	Estimated Unit Cost (LKR)	Estimated Total Cost (LKR)
ESP32 Development Board	1	1,790	1,790
MAX30102 Pulse Oximeter & HR Sensor Module	1	790	790
DS18B20 Temperature Sensor	1	230	230
Buzzer (Active Piezo)	1	50	50
LEDs (Red and Green)	6	20	120
Resistors (220-ohm)	6	15	90
Resistor (4.7k Ohm)	1	60	60
LiPo Battery (3.7V, 1000mAh)	1	850	850
TP4056 LiPo Charger Module	1	170	170
Power Switch	1	100	100
Breadboard	1	260	260
Jumper Wires (Assortment)	1 set	300	300
Micro USB Cable	1	325	325
Plastic Enclosure	1	200	200
Estimated Total Project Cost			**5,335**

All six members of the group will split the estimated total project cost evenly.

8. References

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